

# FELIPE RODRIGUES PEIXOTO DA SILVA

Software Engineer · Full-Stack & AI

São Paulo, Brazil · frps2003@gmail.com · github.com/Comunalti

## SUMMARY

Computer Engineer (Instituto Mauá de Tecnologia, 2025) with 3+ years of industry experience building distributed web platforms and AI pipelines end to end — requirements, architecture, implementation, dockerization and deployment. My work is guided by SOLID, DRY and design patterns, focusing on UI-configurable systems that reduce operational dependency on programmers. Hands-on leadership and people-management experience: student organization president and scholarship teaching assistant for two years, teaching engineering best practices to dozens of students.

## PROFESSIONAL EXPERIENCE

**Full-Stack Developer · Intelicity** Dec 2024 — present

- Co-rewrote the company's core AI image-processing pipeline as a distributed system: containers, orchestration and RabbitMQ messaging.
- Rebuilt the image classification platform with Flutter (client), FastAPI (backend) and MongoDB: parallel image downloads, a single flexible classification mode and 100% GUI-based management.
- Single-handedly built, end to end (deploy, Docker, frontend, backend, database), a platform for city governments to register, browse and audit urban roadwork: users, roles, permissions, per-page/per-feature access control and event-triggered scheduled jobs — all dynamically configurable through the UI.

**Intern — AI & Smart Cities · Intelicity** Mar 2023 — Dec 2024

- Computer vision pipelines in Python: object detection and classification across thousands of images, with georeferenced data in PostgreSQL.
- Created SGC (General Classification System), an internal C# data-annotation tool with per-user authentication, real-time API communication and 4 configurable classification modes — replacing a manual ZIP-based workflow and improving the operations team's process.

## HIGHLIGHTED PROJECTS

**Data-driven software architecture (Thesis)** — A C# engine that interprets rules declared in an external XML file and assembles entities at runtime, with zero recompilation: serialized state machine, declarative parsing and a custom visual editor — the same principles behind configurable enterprise systems.

**Personal full-stack website** — React + Node + TypeScript on Docker Compose, bilingual API-served content, multiple route-based presentation variants and continuous deployment (Vercel + Render) with a custom domain.

## EDUCATION

**B.Sc. in Computer Engineering · Instituto Mauá de Tecnologia** 2021 — 2025

**Mechatronics Technician (integrated with high school) · ETEC Júlio de Mesquita** 2018 — 2020

## ACADEMIC ACTIVITIES

- Student leadership: member (2022) and elected president (2024) of a software development student organization, leading multidisciplinary teams through projects with published deliverables — planning, task allocation and people development.
- Scholarship teaching assistant for two years (2023–2024): taught SOLID, DRY and design patterns (Gang of Four) to freshmen, plus computer graphics and GPGPU lectures — technical communication and raising the code-quality bar.

HARD SKILLS

|                          |                                                                         |
|--------------------------|-------------------------------------------------------------------------|
| Languages                | TypeScript/JavaScript, Python, C#, Dart, SQL                            |
| Backend                  | Node.js, Express, FastAPI, REST APIs, RabbitMQ, WebSockets              |
| Frontend                 | React, Vite, Flutter, HTML/CSS                                          |
| Data                     | PostgreSQL (incl. georeferenced data), MongoDB                          |
| DevOps                   | Docker, Docker Compose, container orchestration, CI/CD, Linux, Git      |
| AI & Vision              | Computer vision, object detection/classification, data annotation       |
| Architecture & Practices | Distributed systems, messaging, RBAC, SOLID, DRY, design patterns (GoF) |
| Other                    | Unity, GPGPU/compute shaders, procedural generation                     |

SOFT SKILLS

People management and team leadership

Technical mentoring and teaching (2 years as TA)

End-to-end ownership (entire platform built solo)

Cross-disciplinary collaboration

Clear technical communication

Pragmatic problem solving

Continuous learning

Product thinking for non-technical users

LANGUAGES

Portuguese — native      English — professional (technical reading, writing and documentation)